

Final Assessment-Alternative Assessment

Semester	202520	Division	CIS
Assessment title in Syllabus	Final_Project_Portfolio	Program	Bachelor of Information Technology/ Bachelor of Information Systems

Course Code	CIS 3603		
Course Title	PROJECT MANAGEMENT		
Course Instructor	Asem Kasem	CRN	22931
Assessment	Project Deliverable 2 – Project Plan	Submission Date	Week 11

Student Declaration:

Academic Integrity Statement

In accordance with the HCT Academic Integrity Policy

- Students are required to refrain from all forms of academic integrity breaches as defined and explained by HCT.
- A student found guilty of having committed acts of academic integrity breach(es) will be subject to the relevant sanctions as outlined by HCT.

إفادة النزاهة الأكاديمية

وفقاً لسياسة كليات التقنية العليا للنزاهة الأكاديمية

- على الطلبة الالتزام بلوائح وقواعد النزاهة الأكاديمية، كما هو مبين وموضح في السياسات والإجراءات الخاصة بكليات التقنية العليا.
- في حالة ارتكاب الطالب أي شكل من أشكال الإخلال بالنزاهة الأكاديمية، سيتعرض إلى العقوبات الموضحة في السياسات ذات الصلة.

This assignment is entirely my/our own work except where I have duly acknowledged other sources in the text and listed those sources at the end of the assignment. I have not previously submitted this work to the HCT, or any other entity. I understand that I may be orally examined on my submission.

Student HCT ID(s):	H00535836	H00495503	H00535463	H00469084	H00
Student Name(s):	Maryam Ali Abdulla	Amna saif obaid	Jawahr Ali Alhosani	Shamsa Saeed Musabeh	
Student Signature					

Chatbot (e.g. ChatGPT) statement

This submission allows you to use AI generated tools, but the prompt used must be stated and the response must be in red font. Include References to ChatGPT.

Instructions

Remove instructions and hints from the template before submitting the final report to BBLearn.



Project Development Approach

Note to students: Select ONE development approach.

All planning documents, terminology, screenshots, and justifications in this report **must align consistently** with the selected approach.

☐ Predictive (Waterfall)

☒ Adaptive (Agile – Scrum)

Planning documents must align to the selected development approach.

Project Plan

Introduction to Project Planning

Project Description

The **FuturePark** project aims to develop an **AI-powered smart parking system** for the Higher Colleges of Technology (HCT). The core deliverable is a **mobile application** that allows students, faculty, and staff to view real-time parking availability, reserve spots, and navigate to available spaces. The system will be supported by **License Plate Recognition (LPR) cameras** and **IoT sensors**, but the focus of this planning document is on the **software development portion** using an **Agile approach**.

Although the overall project follows a **hybrid methodology** (Waterfall for hardware installation and Agile for software development), this report presents the **project management framework** for the **software components** only.

We selected an **Agile approach using Scrum** because it allows us to adapt to changing user needs as students and faculty interact with early versions of the app. Regular **Sprint Reviews** will let HCT management, IT staff, and end users see progress and suggest improvements, ensuring the final product meets their expectations. Delivering working software every Sprint reduces the risk of building the wrong features by validating functionality early.

Project Management Framework: Agile (Scrum)

We have selected an **Agile approach using the Scrum framework**. This methodology is ideal for software projects where requirements may evolve based on user feedback and technological considerations (Schwaber & Sutherland, 2020).

Key Characteristics of Agile/Scrum:

Aspect	Description
Iterative Planning	Work is divided into fixed-length iterations called Sprints (typically 2–4 weeks). At the start of each Sprint, the team plans which features from the Product Backlog will be delivered.
Incremental Delivery	At the end of each Sprint, we deliver a completed and tested set of features that could be released to users if needed. This allows stakeholders to see progress early and provide feedback.
Adaptive Control	The plan is continuously adapted based on feedback from Sprint Reviews and changes in priorities. The Product Backlog is refined throughout the project.
Cadence	Scrum follows a regular cadence: Sprint Planning → Daily Scrum → Sprint Review → Sprint Retrospective . This rhythm ensures frequent inspection and adaptation.

Stakeholder Engagement

Stakeholder Register and Mapping of Influence and Interest

Project Title: FuturePark: AI-Powered Smart Parking System						
Project Start Date : 23- March - 2026						
Project Manager: Maryam Ali Abdulla Bin Hamdah Alali						
Name	Position	Contact Details	Role /Responsibility	Influence	Interest	Engagement Strategy
Dr. Faisal Alayyan	President & CEO, HCT	ceo.office@hct.ac.ae	Executive Sponsor. Provides high-level strategic approval and champions the project across HCT.	High	Medium	Keep Satisfied: Provide a quarterly executive summary highlighting major achievements and alignment with HCT's strategic goals. Involve only in critical, high-level decisions.
Dr. Khaled Al Hammadi	Campus Director, Sharjah	khaled.hammedi@hct.ac.ae	Primary Customer & Stakeholder Lead. Represents the interests of the Sharjah campus. Validates that the solution solves the real-world parking problem.	High	High	Manage Closely: Actively involve him in Sprint Review meetings to get feedback on new features. Schedule monthly check-in meetings to discuss progress and address any campus-specific concerns.
IT Department Head	Head of IT Infrastructure	it.head@hct.ac.ae	Ensures the new system integrates with HCT's network and complies with security policies. Provides technical resources and guidance.	High	High	Manage Closely: The IT Head (or a representative) will be consulted during Sprint Planning for technical feasibility. They will be invited to Sprint Demos to see the product in action and ensure it meets technical standards.
Security Department Head	Head of Campus Security	security.head@hct.ac.ae	Ensures the system enhances	High	Low	Keep Satisfied: Consult with them on the

			campus safety and aligns with security protocols (e.g., gate access, camera monitoring).			design of security-related features (LPR cameras, gate control). Provide them with a system demo once security features are stable. No need to involve them in day-to-day software sprints.
Students & Faculty	End Users	(General)	Use the system daily. Their feedback is crucial for app usability and feature development.	Low	High	Keep Informed: Conduct surveys and focus groups to gather initial requirements. Invite a sample of users to participate in Sprint Reviews for live feedback. Send out monthly email newsletters with updates on new features.
Jawahr Ali Hassan Abdulla Alhosani	Business Analyst (Product Owner)	H00535463@hct.ac.ae	Represents the stakeholders' voice. Manages and prioritizes the Product Backlog to ensure the team is working on the most valuable features.	Medium	High	Manage Closely: Works with the development team daily. Responsible for clarifying requirements and accepting completed user stories at the end of each Sprint.
Amna Saif Obaid Saif Al Ali	Technical Expert (Scrum Master)	H00495503@hct.ac.ae	Facilitates the Scrum process and removes any obstacles blocking the team's progress. Acts as a servant-leader for the team.	Medium	High	Keep Informed & Empowered: Leads Daily Scrums, Sprint Planning, and Retrospectives. Reports on team velocity and progress to the Project Manager.
Shamsa Saeed Musabeh Alnaddas Alteneiji	Developer	H00469084@hct.ac.ae	Designs, develops, and tests the software features as prioritized in the Sprint Backlog.	Low	High	Keep Informed: Participate in all Scrum ceremonies (Daily Scrum, Planning, Review,



						Retrospective) to stay aligned with project goals and report on progress.
Maryam Ali Abdulla Bin Hamdah Alali	Project Manager	H00535836@hct.ac.ae	Leads the project, manages the overall schedule and budget, and serves as the main communication link between the team and all external stakeholders.	High	High	Manage Closely: Oversees the entire project plan and ensures alignment between the team's work and the stakeholders' expectations.

Communication and Engagement Plan

Project Title: FuturePark: AI-Powered Smart Parking System				
Project Start Date : 23- March - 2026				
Project Manager: Maryam Ali Abdulla Bin Hamdah Alali				
Communication Type [Push, Pull, Interactive]	Communication Method [Reports, Daily Standups, Sprint Reviews, Meeting, retrospectives, reviews etc.]	Stakeholders Involved	Frequency of distribution	Distribution method
Interactive	Daily Scrum (Standup)	Project Team (Maryam, Amna, Jawahr, Shamsa)	Daily (at the start of each workday)	In-person, 15-minute meeting at the project board.
Interactive	Sprint Planning Meeting	Project Team (Maryam, Amna, Jawahr, Shamsa)	Bi-weekly (at the beginning of each new Sprint)	2-hour workshop to plan the upcoming Sprint's work.
Interactive	Sprint Review Meeting	Project Team, Dr. Khaled Al Hammadi, Dr. Asem Kasem, IT Dept. Head, (End User representatives invited)	Bi-weekly (at the end of each Sprint)	1-hour formal meeting to demonstrate the completed features and gather feedback. (In-person & MS Teams).
Interactive	Sprint Retrospective	Project Team (Maryam, Amna, Jawahr, Shamsa)	Bi-weekly (after the Sprint Review)	1-hour internal team meeting to discuss what went well and what can be improved.
Push	Weekly Progress Update	Dr. Asem Kasem, Dr. Khaled Al Hammadi	Weekly (every Thursday)	Email with a brief summary of Sprint progress, key achievements, and any immediate risks.
Push	Monthly Executive Summary	Dr. Faisal Alayyan	Monthly	Formal one-page report (PDF) sent via email, highlighting major milestones and alignment with HCT's strategic goals.
Push	Meeting Minutes & Action Items	All meeting attendees	Within 24 hours after any formal meeting (e.g., Sprint Review)	Email summary.



Pull	Project Documentation (Product Backlog, Risk Register, etc.)	All Stakeholders (as needed)	Continuously updated	Stored on a shared platform like Microsoft Teams/SharePoint for on-demand access (Schwaber & Sutherland, 2020).
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Project Scope

Scope documents

(This section must have filled in documents for either the Waterfall or Agile development approaches. Use one or the other)

Waterfall Documents	Agile Documents
Scope Statement	Product Vision
Product Scope Inclusions	User stories Product Backlog with prioritization
Exclusions	Out-of-scope backlog items
WBS by Phase	WBS by Iterations of 2 weeks or Increments

Project Scope

In our Agile approach for the **FuturePark** project, scope management is treated as a dynamic and ongoing process, rather than a one-time event defined at the start. The project's direction is guided by a clear **Product Vision**, which sets the long-term goal. The detailed scope is managed through the **Product Backlog**, a living document that contains all desired features and requirements.

Requirements are captured as **User Stories**, which allows the team to prioritize them based on the value they deliver to stakeholders. At the beginning of each **Sprint**, the team selects a number of high-priority user stories to work on, forming the **Sprint Backlog**. This iterative and incremental approach allows for continuous adaptation to feedback, ensuring that we build the most important features first and deliver a product that truly meets the needs of the HCT community.

Product Vision

For HCT students, staff, and faculty who struggle with the daily stress and wasted time of finding a parking space, the FuturePark system is a smart, AI-powered parking solution that provides real-time parking guidance and advance reservations through a seamless mobile app.

Unlike the current manual, inefficient, and frustrating parking process, our product delivers a modern, reliable, and convenient experience that saves valuable time, reduces campus congestion, and reflects HCT's commitment to innovation.

User Stories

Core Functionality:

1. **As a student, I want to** view a real-time map of available parking spots **so that** I can avoid wasting time searching for a space.
2. **As a new user, I want to** create a profile and register my vehicle's license plate **so that** the system can identify me for automatic access.
3. **As a faculty member, I want to** reserve a parking spot in advance **so that** I can guarantee a space is available before an important lecture.
4. **As a driver, I want to** receive turn-by-turn navigation to my parking spot **so that** I can find it quickly and easily.

Access and Automation:

5. **As a** registered user, **I want** the parking gate to open automatically **so that** I can enter and exit smoothly without needing a physical card.
6. **As a** campus visitor, **I want to** generate a temporary QR code for entry **so that** I can access the parking facilities for a single day.

Notifications and Management:

7. **As a** user with a reservation, **I want to** receive a push notification before my booking ends **so that** I have time to move my car or extend my session.
8. **As a** user, **I want to** view my complete parking history in the app **so that** I can track my time on campus.

Administrative Functionality:

9. **As a** Campus Director, **I want to** access a dashboard with real-time analytics **so that** I can make data driven decisions about parking management and future needs.
10. **As a** Security Officer, **I want to** have a manual override for the gates **so that** I can control access during emergencies or system issues.
- 11.

Product Backlog with prioritization

ID	As a...	I want to...	So that...	Priority	Sprint	Status
1	new user	create a profile and register my vehicle's license plate	the system can identify me for automatic access.	Must	1	To be started
2	student	view a real-time map of available parking spots	I can avoid wasting time searching for a space.	Must	1	To be started
3	registered user	have the parking gate open automatically	I can enter and exit smoothly without needing a physical card.	Must	1	To be started
4	faculty member	reserve a parking spot in advance	I can guarantee a space is available before an important lecture.	Should	2	To be started
5	driver	receive turn-by-turn navigation to my parking spot	I can find it quickly and easily.	Should	2	To be started
6	Campus Director	access a dashboard with real-time analytics	I can make data-driven decisions about parking management.	Should	2	To be started
7	user with a reservation	receive a push notification before my booking ends	I have time to move my car or extend my session.	Could	3	To be started
8	user	view my complete parking history in the app	I can track my time on campus.	Could	3	To be started

9	campus visitor	generate a temporary QR code for entry	I can access the parking facilities for a single day.	Could	3	To be started
10	Security Officer	have a manual override for the gates	I can control access during emergencies or system issues.	Must	3	To be started

Out-of-scope backlog items

The features listed below were discussed but are not included in this version of FuturePark. We are focusing on the most important features first to deliver the project on time. These ideas can be considered for future updates.

1- "Find My Car" Feature:

- The app will not include a feature that uses GPS or indoor positioning to help users locate where they parked their car. Users are responsible for remembering their parking spot.

2- Electric Vehicle (EV) Charging Station Management:

- The system will not integrate with EV charging stations to show their availability, manage booking, or handle payments for charging. This can be a separate future project.

3- Advanced User Analytics and Personalized Recommendations:

- While the admin dashboard will show general usage patterns, the app will not provide personalized analytics to users (e.g., "You usually park in Zone B") or recommend "best" spots based on individual habits.

4- Integration with Public Transport and Carpooling Services:

- The app will not connect to local public transport systems (like bus schedules) or integrate with carpooling platforms to help users coordinate rides.

5- Multi-Campus Support:

- The initial release of the system will be deployed and tested only at the **Sharjah Campus**. The functionality to view or book parking at other HCT campuses is out of scope for this project phase.

WBS by Iterations of 2 weeks or Increments

At the start of each Sprint, we break down high-priority User Stories from the Product Backlog into smaller, actionable tasks.

The tables below show the plan for Sprint 1 and Sprint 2, including the tasks and effort estimates for each. This Work Breakdown Structure (WBS) represents the high-level deliverables; detailed tasks will be managed and tracked within the Product Backlog during each Sprint.

Sprint 1 (Weeks 1-2)

Sprint Goal: To deliver the core functionality allowing users to register, see the live parking map, and gain automatic entry.

ID	Work Item (Task)	Belongs to User Story (ID)	Estimated Effort
1.1	Design UI for user registration and profile screens.	1	2 days
1.2	Develop backend API for creating and managing user accounts.	1	3 days
1.3	Connect registration UI with the backend API.	1	1 day
1.4	Test the complete registration and login process.	1	1 day
2.1	Design the real-time map interface in the mobile app.	2	1 day
2.2	Develop a service to receive and process data from parking sensors.	2	3 days
2.3	Display parking availability status (free/occupied) on the map UI.	2	2 days
3.1	Integrate with the License Plate Recognition (LPR) camera system.	3	3 days
3.2	Develop the logic to match a license plate to a registered user and open the gate.	3	2 days
3.3	Conduct end-to-end testing of the automatic gate entry.	3	2 days

Sprint 2 (Weeks 3-4)

Sprint Goal: To implement the parking reservation feature and the administrative dashboard

ID	Work Item (Task)	Belongs to User Story (ID)	Estimated Effort
4.1	Design the UI for selecting a date/time and reserving a parking spot.	4	2 days
4.2	Develop backend logic to manage bookings, availability, and prevent double-booking.	4	3 days
4.3	Test the complete reservation workflow from booking to arrival.	4	2 days
5.1	Design the UI for the in-app navigation feature.	5	1 day
5.2	Integrate with a mapping service (e.g., Google Maps API) to provide directions.	5	2 days
6.1	Design the UI for the admin dashboard showing key metrics (occupancy, peak times).	6	2 days
6.2	Develop backend APIs to aggregate and provide data for the dashboard.	6	3 days
6.3	Test the accuracy and real-time updates of the dashboard.	6	1 day

WBS Outline or Product Backlog created in MS Project

Work Breakdown Structure (WBS)

The project was structured using a Work Breakdown Structure in MS Project by dividing the system into three sprints (Stafford, 2019).

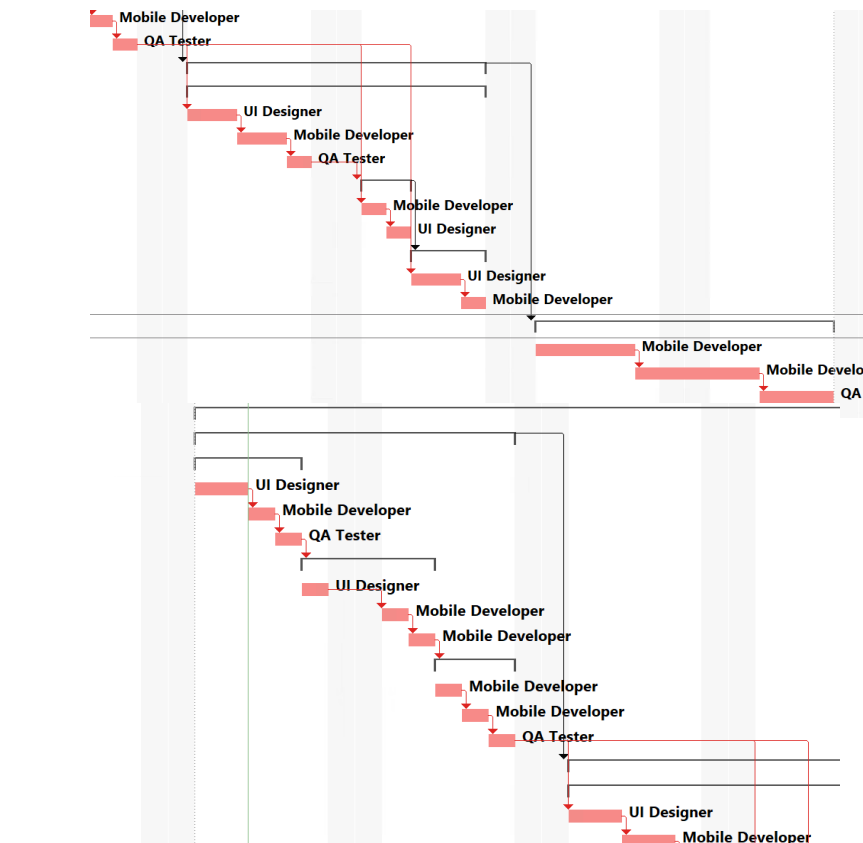
Each sprint includes modules and detailed tasks that were logically sequenced using task dependencies to ensure smooth execution and proper timeline control.

i	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
		FuturePark Mobile Application	30 days	Mon 3/23/26	Fri 5/1/26		Product Owner
		Sprint 1 – Core Features	10 days	Mon 3/23/26	Fri 4/3/26		Scrum Master
		User Registration Module	4 days	Mon 3/23/26	Thu 3/26/26		
		Design Registration UI	2 days	Mon 3/23/26	Tue 3/24/26		UI Designer
		Develop Authentication API	1 day	Wed 3/25/26	Wed 3/25/26	4	Mobile Developer
		Testing Registration	1 day	Thu 3/26/26	Thu 3/26/26	5	QA Tester
		Real Time Parking Map	3 days	Fri 3/27/26	Tue 3/31/26	6	
		Design Map UI	1 day	Fri 3/27/26	Fri 3/27/26		UI Designer
		Integrate Sensor Data	1 day	Mon 3/30/26	Mon 3/30/26	8	Mobile Developer
		Display Availability	1 day	Tue 3/31/26	Tue 3/31/26	9	Mobile Developer
		Gate Automation	3 days	Wed 4/1/26	Fri 4/3/26	10	
		Integrate LPR Camera	1 day	Wed 4/1/26	Wed 4/1/26		Mobile Developer
		License Matching Logic	1 day	Thu 4/2/26	Thu 4/2/26	12	Mobile Developer
		End to End testing	1 day	Fri 4/3/26	Fri 4/3/26	13	QA Tester
		Sprint 2 – Reservation Features	10 days	Mon 4/6/26	Fri 4/17/26	2	Scrum Master
		Reserve Spot Module	10 days	Mon 4/6/26	Fri 4/17/26		
		Design Reservation UI	2 days	Mon 4/6/26	Tue 4/7/26	14	UI Designer
		Develop Booking Logic	2 days	Wed 4/8/26	Thu 4/9/26	17	Mobile Developer
		Test Reservation	1 day	Fri 4/10/26	Fri 4/10/26	18	QA Tester
		Navigation	2 days	Mon 4/13/26	Tue 4/14/26	19	
		Integrate Maps API	1 day	Mon 4/13/26	Mon 4/13/26	14	Mobile Developer
		Design Navigation UI	1 day	Tue 4/14/26	Tue 4/14/26	21	UI Designer
		Admin Dashboard	3 days	Wed 4/15/26	Fri 4/17/26	20	
		Design Dashboard	2 days	Wed 4/15/26	Thu 4/16/26	14	UI Designer
		Develop Analytics API	1 day	Fri 4/17/26	Fri 4/17/26	24	Mobile Developer
		Sprint 3 – Enhancement Features	10 days	Mon 4/20/26	Fri 5/1/26	15	Scrum Master
		Push Notifications	4 days	Mon 4/20/26	Thu 4/23/26		Mobile Developer
		Parking History	3 days	Fri 4/24/26	Tue 4/28/26	27	Mobile Developer
		Manual Override	3 days	Wed 4/29/26	Fri 5/1/26	28	QA Tester

Network Diagram

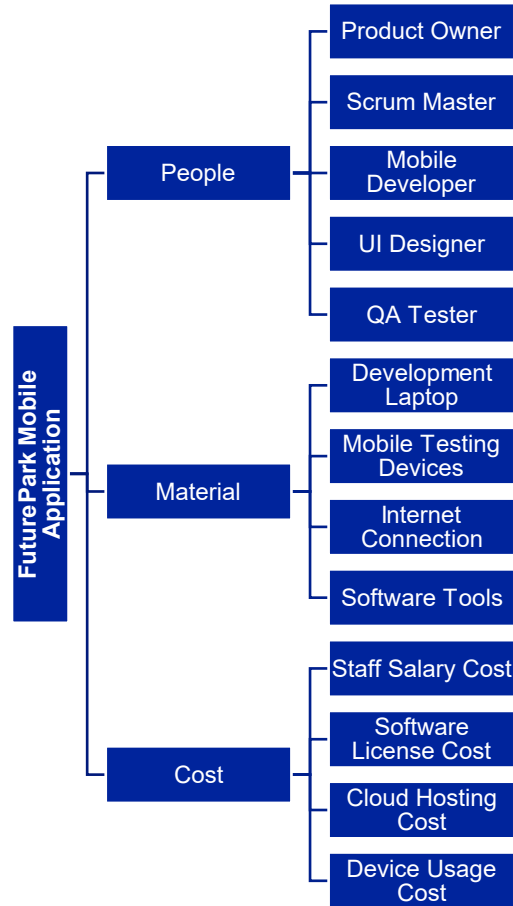
The Network Diagram illustrates the relationship between project tasks and their dependencies.

This view helps identify the critical path and ensures that tasks are executed in the correct order to avoid project delays



Project Resource Plan

Resource breakdown Structure



Resource Breakdown Structure

A Resource Breakdown Structure was developed to organize project resources into three main categories: people, materials, and costs.

This hierarchical structure supports better resource planning and improves visibility of required project inputs.

Resource Sheet and Resource Assignment in a Project Management Software

Project Cost Analysis

The project cost table shows the estimated total cost for each sprint and task. Costs were calculated automatically based on resource hourly rates and task durations, enabling effective budget estimation and cost monitoring.

Resource Name	Type	Material Label	Init	Grot	Max. Units	Std. Rate	Ovt. Rate	Cost/Use	Accrue At
Product Owner	Work		P		100%	300.00 ج.د./hr	0.00 ج.د./hr	0.00 ج.د.	Prorated
Scrum Master	Work		S		100%	250.00 ج.د./hr	0.00 ج.د./hr	0.00 ج.د.	Prorated
Mobile Developer	Work		M		100%	220.00 ج.د./hr	0.00 ج.د./hr	0.00 ج.د.	Prorated
UI Designer	Work		U		100%	180.00 ج.د./hr	0.00 ج.د./hr	0.00 ج.د.	Prorated
QA Tester	Work		Q		100%	150.00 ج.د./hr	0.00 ج.د./hr	0.00 ج.د.	Prorated
Development Laptop	Material	Unit	D			4,000.00 ج.د.		0.00 ج.د.	Prorated
Mobile Testing Devices	Material	Device	M			2,500.00 ج.د.		0.00 ج.د.	Prorated
Internet Connection	Material	m.	I			300.00 ج.د.		0.00 ج.د.	Prorated
Software Tools	Material	license	S			800.00 ج.د.		0.00 ج.د.	Prorated
Staff Salary Cost	Cost		S						Prorated
Cloud Hosting Cost	Cost		C						Prorated

Resource Name	Cost	Baseline Cost	Variance	Actual Cost	Remaining	Add New Column
Product Owner	72,000.00 ج.د.	0.00 ج.د.	72,000.00 ج.د.	0.00 ج.د.	72,000.00 ج.د.	
Scrum Master	60,000.00 ج.د.	0.00 ج.د.	60,000.00 ج.د.	0.00 ج.د.	60,000.00 ج.د.	
Mobile Developer	28,160.00 ج.د.	0.00 ج.د.	28,160.00 ج.د.	0.00 ج.د.	28,160.00 ج.د.	
UI Designer	11,520.00 ج.د.	0.00 ج.د.	11,520.00 ج.د.	0.00 ج.د.	11,520.00 ج.د.	
QA Tester	7,200.00 ج.د.	0.00 ج.د.	7,200.00 ج.د.	0.00 ج.د.	7,200.00 ج.د.	
Development Laptop	0.00 ج.د.	0.00 ج.د.	0.00 ج.د.	0.00 ج.د.	0.00 ج.د.	
Mobile Testing Devices	0.00 ج.د.	0.00 ج.د.	0.00 ج.د.	0.00 ج.د.	0.00 ج.د.	
Internet Connection	0.00 ج.د.	0.00 ج.د.	0.00 ج.د.	0.00 ج.د.	0.00 ج.د.	
Software Tools	0.00 ج.د.	0.00 ج.د.	0.00 ج.د.	0.00 ج.د.	0.00 ج.د.	
Staff Salary Cost	15,000.00 ج.د.	0.00 ج.د.	15,000.00 ج.د.	0.00 ج.د.	15,000.00 ج.د.	
Cloud Hosting Cost	28,500.00 ج.د.	0.00 ج.د.	28,500.00 ج.د.	0.00 ج.د.	28,500.00 ج.د.	

Task Name	Fixed Cost	Accrual	Total Cost	Baseline	Variance	Actual	Remaining
1 FuturePark Mobile Application	0.00 ج.د.	Prorated	222,380.00 ج.د.	222,380.00 ج.د.	0.00 ج.د.	0.00 ج.د.	222,380.00 ج.د.
2 Sprint 1 – Core Features	0.00 ج.د.	Prorated	42,520.00 ج.د.	42,520.00 ج.د.	0.00 ج.د.	0.00 ج.د.	42,520.00 ج.د.
3 User Registration Module	0.00 ج.د.	Prorated	7,840.00 ج.د.	7,840.00 ج.د.	0.00 ج.د.	0.00 ج.د.	7,840.00 ج.د.
4 Design Registration	0.00 ج.د.	Prorated	2,880.00 ج.د.	2,880.00 ج.د.	0.00 ج.د.	0.00 ج.د.	2,880.00 ج.د.
5 Develop Authentication	0.00 ج.د.	Prorated	3,760.00 ج.د.	3,760.00 ج.د.	0.00 ج.د.	0.00 ج.د.	3,760.00 ج.د.
6 Testing Registration	0.00 ج.د.	Prorated	1,200.00 ج.د.	1,200.00 ج.د.	0.00 ج.د.	0.00 ج.د.	1,200.00 ج.د.
7 Real Time Parking Map	0.00 ج.د.	Prorated	4,960.00 ج.د.	4,960.00 ج.د.	0.00 ج.د.	0.00 ج.د.	4,960.00 ج.د.
8 Design Map UI	0.00 ج.د.	Prorated	1,440.00 ج.د.	1,440.00 ج.د.	0.00 ج.د.	0.00 ج.د.	1,440.00 ج.د.
9 Integrate Sensor Data	0.00 ج.د.	Prorated	1,760.00 ج.د.	1,760.00 ج.د.	0.00 ج.د.	0.00 ج.د.	1,760.00 ج.د.
10 Display Availability	0.00 ج.د.	Prorated	1,760.00 ج.د.	1,760.00 ج.د.	0.00 ج.د.	0.00 ج.د.	1,760.00 ج.د.
11 Gate Automation	0.00 ج.د.	Prorated	4,720.00 ج.د.	4,720.00 ج.د.	0.00 ج.د.	0.00 ج.د.	4,720.00 ج.د.
12 Integrate LPR Camera	0.00 ج.د.	Prorated	1,760.00 ج.د.	1,760.00 ج.د.	0.00 ج.د.	0.00 ج.د.	1,760.00 ج.د.
13 License Matching Logic	0.00 ج.د.	Prorated	1,760.00 ج.د.	1,760.00 ج.د.	0.00 ج.د.	0.00 ج.د.	1,760.00 ج.د.
14 End to End testing	0.00 ج.د.	Prorated	1,200.00 ج.د.	1,200.00 ج.د.	0.00 ج.د.	0.00 ج.د.	1,200.00 ج.د.
Sprint 2 – Reservation Features	0.00 ج.د.	Prorated	65,440.00 ج.د.	65,440.00 ج.د.	0.00 ج.د.	0.00 ج.د.	65,440.00 ج.د.
Reserve Spot Module	0.00 ج.د.	Prorated	40,440.00 ج.د.	40,440.00 ج.د.	0.00 ج.د.	0.00 ج.د.	40,440.00 ج.د.
Design Reservation	0.00 ج.د.	Prorated	2,880.00 ج.د.	2,880.00 ج.د.	0.00 ج.د.	0.00 ج.د.	2,880.00 ج.د.
Develop Booking Logic	0.00 ج.د.	Prorated	28,520.00 ج.د.	28,520.00 ج.د.	0.00 ج.د.	0.00 ج.د.	28,520.00 ج.د.
Test Reservation	0.00 ج.د.	Prorated	1,200.00 ج.د.	1,200.00 ج.د.	0.00 ج.د.	0.00 ج.د.	1,200.00 ج.د.
Navigation	0.00 ج.د.	Prorated	3,200.00 ج.د.	3,200.00 ج.د.	0.00 ج.د.	0.00 ج.د.	3,200.00 ج.د.
Integrate Maps API	0.00 ج.د.	Prorated	1,760.00 ج.د.	1,760.00 ج.د.	0.00 ج.د.	0.00 ج.د.	1,760.00 ج.د.
Design Navigation	0.00 ج.د.	Prorated	1,440.00 ج.د.	1,440.00 ج.د.	0.00 ج.د.	0.00 ج.د.	1,440.00 ج.د.
Admin Dashboard	0.00 ج.د.	Prorated	4,640.00 ج.د.	4,640.00 ج.د.	0.00 ج.د.	0.00 ج.د.	4,640.00 ج.د.
Design Dashboard	0.00 ج.د.	Prorated	2,880.00 ج.د.	2,880.00 ج.د.	0.00 ج.د.	0.00 ج.د.	2,880.00 ج.د.
Develop Analytics	0.00 ج.د.	Prorated	1,760.00 ج.د.	1,760.00 ج.د.	0.00 ج.د.	0.00 ج.د.	1,760.00 ج.د.
Sprint 3 – Enhancements	0.00 ج.د.	Prorated	42,420.00 ج.د.	42,420.00 ج.د.	0.00 ج.د.	0.00 ج.د.	42,420.00 ج.د.
Push Notifications	0.00 ج.د.	Prorated	8,540.00 ج.د.	8,540.00 ج.د.	0.00 ج.د.	0.00 ج.د.	8,540.00 ج.د.
Parking History	0.00 ج.د.	Prorated	5,280.00 ج.د.	5,280.00 ج.د.	0.00 ج.د.	0.00 ج.د.	5,280.00 ج.د.
Manual Override	0.00 ج.د.	Prorated	3,600.00 ج.د.	3,600.00 ج.د.	0.00 ج.د.	0.00 ج.د.	3,600.00 ج.د.

Image 4 — Project Cost Analysis

The project cost table shows the estimated total cost for each sprint and task. Costs were calculated automatically based on resource hourly rates and task durations, enabling effective budget estimation and cost monitoring.

Schedule Planning and Estimation

Duration

To estimate the duration of the major task we have used the PERT technique . By calculating task duration based on the optimistic, most likely, and pessimistic values to ensure that the assigned story points are realistic and aligned with the expected effort (Project Management Institute, 2021).

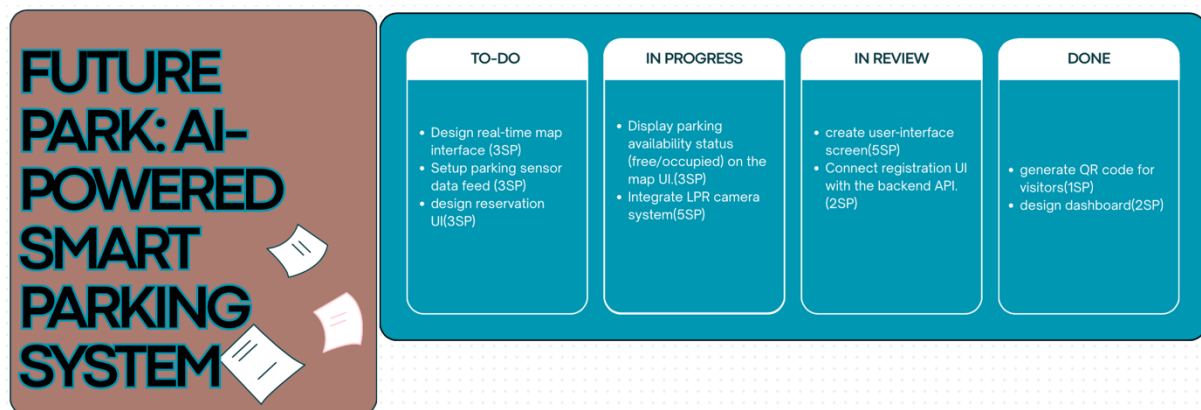
The chosen major task to apply the PERT technique to is the sub task 1.2 Develop authentication API for user registration in sprint 1.

Calculation: $PERT = (Optimistic + 4 \times Most\ Likely + Pessimistic) \div 6$
 $= (2 + 4 \times 3 + 4) / 6 = 3$

PERT = 3 days

This confirms that the task should be estimated at 3 story point (fibonacci) meaning it's at a moderate level where one developer could finish the task in a 10 day sprint.

In this part we used the scrum board to track tasks during sprints such as design reservation, displaying g parking availability status and generating a part time QR code for visitors. The Tasks are organized in the To Do, In Progress, In review and Done stages where each task is assigned to a story points using the Fibonacci scale (1, 2, 3, 5) to estimate the effort. for example the task "Design real-time map interface" is assigned to 3 story point, meaning the effort of doing this task is moderate, isn't easy neither complex and that's how the scrum board is used.

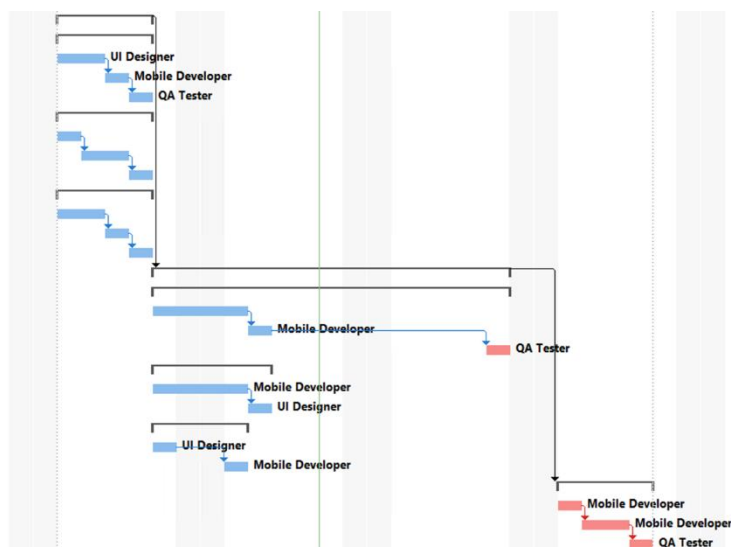


Project Schedule created using a PM Software

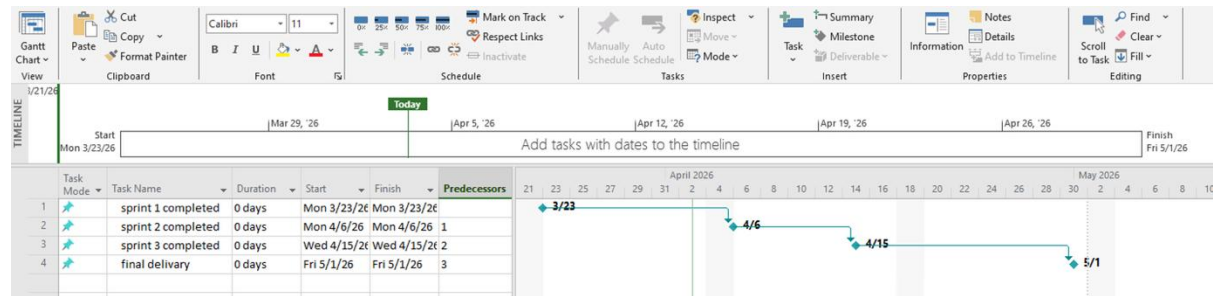
Task List

WBS	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
1		FuturePark Mobile Application	19 days	Mon 3/23/26	Thu 4/16/26		Product Owner
1.1		Sprint 1 – Core Features	4 days	Mon 3/23/26	Thu 3/26/26		Scrum Master
1.1.1		User Registration Module	4 days	Mon 3/23/26	Thu 3/26/26		
1.1.1.1		Design Registration UI	2 days	Mon 3/23/26	Tue 3/24/26		UI Designer
1.1.1.2		Develop Authentication API	1 day	Wed 3/25/26	Wed 3/25/26	4	Mobile Develop
1.1.1.3		Testing Registration	1 day	Thu 3/26/26	Thu 3/26/26	5	QA Tester
1.1.2		Real Time Parking Map	4 days	Mon 3/23/26	Thu 3/26/26		
1.1.2.1		Design Map UI	1 day	Mon 3/23/26	Mon 3/23/26		
1.1.2.2		Integrate Sensor Data	2 days	Tue 3/24/26	Wed 3/25/26	8	
1.1.2.3		Display Availability	1 day	Thu 3/26/26	Thu 3/26/26	9	
1.1.2.4		Gate Automation	4 days	Mon 3/23/26	Thu 3/26/26		
1.1.2.4.1		Integrate LPR Camera	2 days	Mon 3/23/26	Tue 3/24/26		
1.1.2.4.2		License Matching Logic	1 day	Wed 3/25/26	Wed 3/25/26	12	
1.1.2.4.3		End to End testing	1 day	Thu 3/26/26	Thu 3/26/26	13	
1.2		Sprint 2 – Reservation Features	11 days	Fri 3/27/26	Fri 4/10/26	2	UI Designer
1.2.1		Reserve Spot Module	11 days	Fri 3/27/26	Fri 4/10/26		
1.2.1.1		Design Reservation UI	2 days	Fri 3/27/26	Mon 3/30/26		
1.2.1.2		Develop Booking Logic	1 day	Tue 3/31/26	Tue 3/31/26	17	Mobile Develop
1.2.1.3		Test Reservation	1 day	Fri 4/10/26	Fri 4/10/26	18	QA Tester
1.2.2		Navigation	3 days	Fri 3/27/26	Tue 3/31/26		
1.2.2.1		Integrate Maps API	2 days	Fri 3/27/26	Mon 3/30/26		Mobile Develop
1.2.2.2		Design Navigation UI	1 day	Tue 3/31/26	Tue 3/31/26	21	UI Designer
1.2.3		Admin Dashboard	2 days	Fri 3/27/26	Mon 3/30/26		
1.2.3.1		Design Dashboard	1 day	Fri 3/27/26	Fri 3/27/26		UI Designer
1.2.3.2		Develop Analytics API	1 day	Mon 3/30/26	Mon 3/30/26	24	Mobile Develop
1.3		Sprint 3 – Enhancement Features	4 days	Mon 4/13/26	Thu 4/16/26	15	
1.3.1		Push Notifications	1 day	Mon 4/13/26	Mon 4/13/26		Mobile Develop
1.3		Sprint 3 – Enhancement Features	4 days	Mon 4/13/26	Thu 4/16/26	15	
1.3.1		Push Notifications	1 day	Mon 4/13/26	Mon 4/13/26		Mobile Develop
1.3.2		Parking History	2 days	Tue 4/14/26	Wed 4/15/26	27	Mobile Develop
1.3.3		Manual Override	1 day	Thu 4/16/26	Thu 4/16/26	28	QA Tester

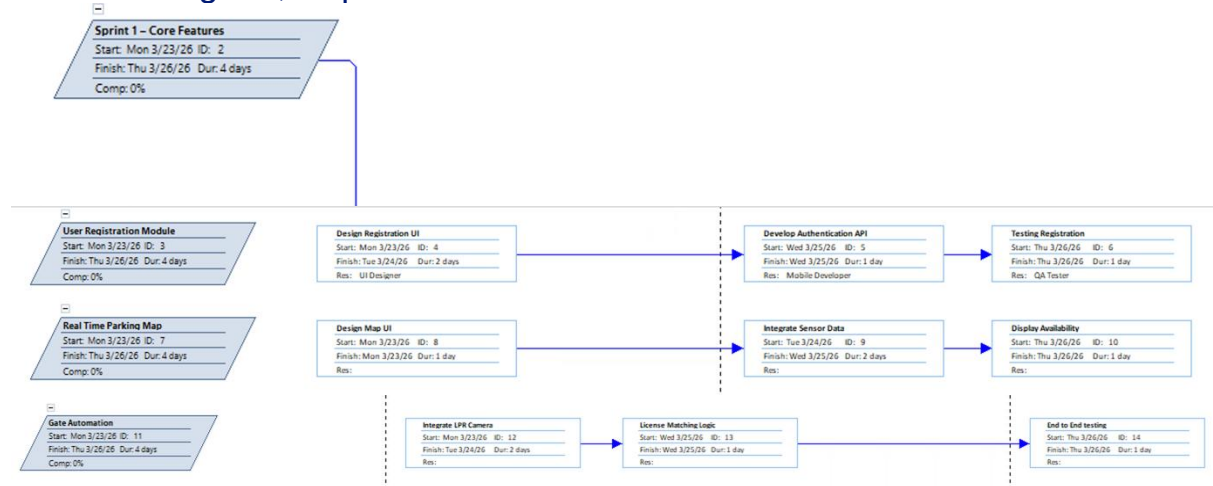
Gantt Chart



Milestone List



Network Diagram, Dependencies and Critical Paths



Finding the critical path for path A, B AND C in sprint 1:

Path A (user registration): $2+1+1=4$ days

Path B (Real time parking map): $1+2+1=4$ days

Path C (Gate Automation): $2+1+1=4$ days

Critical Path for Sprint 1:

I chose to calculate the critical path for the sub tasks in sprint 1, so based on the network diagram, the sprint contains three parallel subtasks:

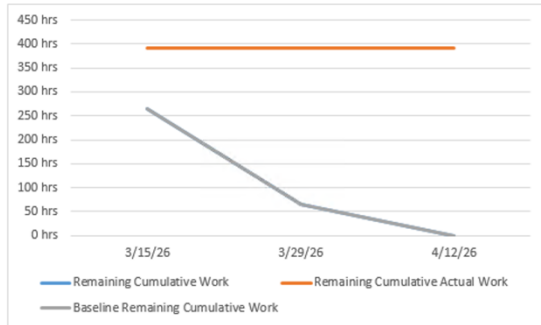
- User Registration Module (4 days)
- Real Time Parking Map (4 days)
- Gate Automation (4 days)

All subtasks start on March 23 and finish on March 26 therefore they run in parallel with no internal dependencies including the critical path duration being 4 days. This means any delay in any of these subtasks will delay the entire Sprint 1.

Total Sprint 1 duration = 4 days

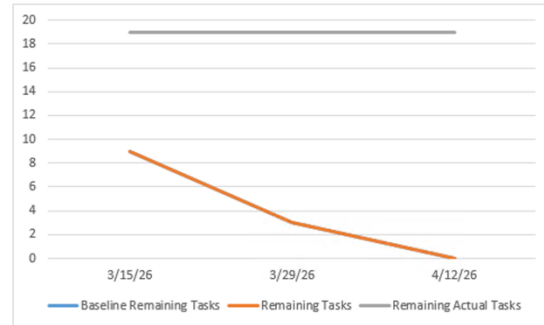
Mon 3/23/26 - Thu 4/16/26

BURNDOWN



WORK BURNDOWN

Shows how much work you have completed and how much you have left. If the remaining cumulative work line is steeper, then the project may be late. Is your baseline zero?



TASK BURNDOWN

Shows how many tasks you have completed and how many you have left. If the remaining tasks line is steeper, then your project may be late.

Cost/Budget Planning and Estimation

Estimation techniques used to determine budget

The use of Agile estimation methods was central to preparing a budget for FuturePark. Bottom-up estimation was performed by breaking user stories into smaller tasks within each Sprint and assigning story points based on complexity. The story points were then converted into hours of estimated effort, which were multiplied by the resource hourly rate to obtain a cost estimate.

Parametric estimation was used by applying a fixed cost-per-story point approach. The team determined that one story point could be assumed to equal approximately 6 hours of work and that the average hourly wage for team members was \$20. Therefore, the overall budget for the project could be estimated at \$120 per story point (6 hours x \$20 = \$120).

Both methods produced reliable and flexible estimates that were consistent with Agile development methodologies (Project Management Institute, 2021).

Budgeting using a PM Software

Budget Calculation Table

The table below shows the estimated cost of key project features based on assigned story points:

Task	Story Points	Cost per Story Point (\$120)	Total Cost (\$)
User Registration Module	3	120	360
RealTime Parking Map	3	120	360
Gate Automation System	5	120	600
Parking Reservation Feature	5	120	600
Navigation Integration	2	120	240
Admin Dashboard	3	120	360

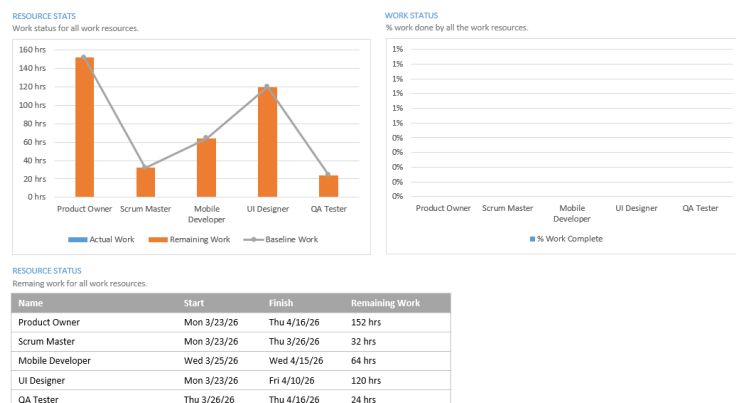
Total Estimated Cost = \$2,520

Justification of Estimation

The budget estimation is based on story points, which reflect the effort required for each task. By converting story points into hours and applying an average hourly rate, the project ensures a flexible and scalable cost estimation. This method aligns with Agile practices and allows adjustments as project requirements evolve.

Resource Name	Type	Material	Initials	Group	Max.	Std. Rate	Ovt. Rate	Cost/Use	Accrue	Base Calendar	Code
Product Owner	Work		P		100%	\$25.00/hr	\$0.00/hr	\$0.00	Prorated	Standard	
Scrum Master	Work		S		100%	\$22.00/hr	\$0.00/hr	\$0.00	Prorated	Standard	
Mobile Developer	Work		M		100%	\$20.00/hr	\$0.00/hr	\$0.00	Prorated	Standard	
UI Designer	Work		U		100%	\$18.00/hr	\$0.00/hr	\$0.00	Prorated	Standard	
QA Tester	Work		Q		100%	\$15.00/hr	\$0.00/hr	\$0.00	Prorated	Standard	

RESOURCE OVERVIEW



Risk Planning

Risk Identification and Analysis

Project Title: FuturePark: AI-Powered Smart Parking System							
Project Start Date: 23-March-2026							
Project Manager: Maryam Ali Abdulla Bin Hamdah Alali							
No.	Risk	Description	Category	Risk Owner	Probability	Impact	Risk Score
1	Changing Requirements	Frequent changes in user requirements during sprints may delay development	Scope	Product Owner	4	4	16
2	Integration Issues	Difficulty integrating mobile app with LPR cameras and IoT sensors	Technical	Developer	3	5	15
3	Resource Availability	The human resource that is needed to complete the project not available.	Resource	Project Manager	3	4	12
4	Technology Failure	Failure of APIs or external services like maps or navigation	Technical	Developer	2	5	10
5	Schedule Delay	Tasks may take longer than estimated causing sprint delays	Schedule	Scrum Master	3	4	12

Risk Response Strategy

No	Response Strategy	Mitigation
1	Mitigation	Regular backlog refinement and stakeholder involvement will help manage changing requirements and reduce disruptions.
2	Mitigation	Early integration and testing of LPR cameras and IoT systems will minimize technical risks during development.
3	Mitigation	Proper task planning and workload distribution will ensure resource availability and handle unexpected absences.
4	Contingency	Backup APIs and alternative services will be prepared to handle any technology failures.
5	Mitigation	Sprint planning and daily stand-ups will help track progress and address delays early.

Most of the FuturePark project's risk response strategies focus on how to deal with identified risks (e.g., changing requirements, inability to integrate all the necessary elements properly, availability of resources, and schedule delays) by mitigating risks through utilizing specific Agile practices (e.g., continuously refining the product backlog, performing early testing on deliverables, conducting appropriate task

planning, and performing regular sprint reviews), to decrease the likelihood or impact of these risk factors during development. For technical failures, contingency methods will be used by creating backup solutions, such as using different applications programming interfaces (APIs), in order to continue the system's operations. The overall effect of these responses is to provide a level of stability for the project (Project Management Institute, 2021), while allowing for successful completion within the established timeline.

End of Planning Document.

References:

Schwaber, K., & Sutherland, J. (2020). *The Scrum Guide*. Scrum.org.
<https://scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf>

Project Management Institute. (2021). *A guide to the project management body of knowledge (PMBOK guide) (7th ed.)*. <https://www.pmi.org/pmbok-guide-standards/foundational/pmbok>

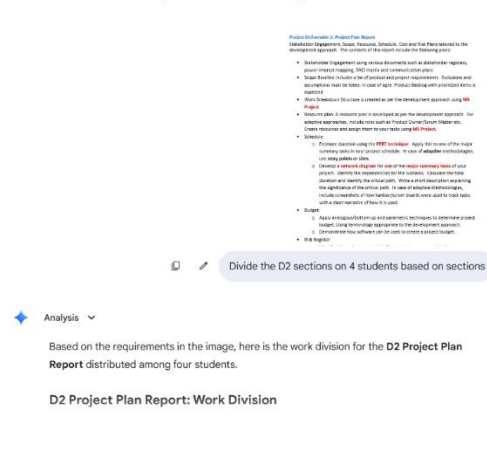
Stafford, B. (2019). *Microsoft Project 2019 and Project Online Desktop Essential Training*. LinkedIn Learning. <https://www.linkedin.com/learning/microsoft-project-2019-and-project-online-desktop-essential-training>

Attachment:

FuturePark_D2_Plan_H00535836_H00535463_H00469084_H00495503.mpp

Declaration on the use of AI (can be a separate document)

The prompt used for task allocation were generated using Gemini (Google AI) on March 22, 2026. The prompt used in this case is as follows: Divide the D2 sections on 4 students based on sections.



Divide the D2 sections on 4 students based on sections

Analysis

Based on the requirements in the image, here is the work division for the **D2 Project Plan Report** distributed among four students.

D2 Project Plan Report: Work Division

Student	Assigned Sections & Key Tasks		
Student 1	Stakeholder Engagement & Scope Baseline * Create Stakeholder Register, Power-Interest Mapping, RACI Matrix, and Communication Plan. * List Products/Project requirements, exclusions, and assumptions (or Product Backlog).		
Student 2	Work Breakdown Structure (WBS) & Resource Plan * Build the WBS using MS Project based on your development approach. * Define roles (e.g., Scrum Master) and assign resources to tasks within MS Project.		
Student 3	Project Schedule * Estimate durations using the PERT technique (or story points). * Develop a Network Diagram for a major task, identify dependencies/critical path, and provide a narrative.		
Student 4	Budget & Risk Register * Apply budgeting techniques (Bottom-up/Parametric) and demonstrate software use. * Identify/categorize risks, perform qualitative analysis, and develop response strategies.		

End of Documentation.